

INFO 465

AI/ML Business Case – Incident Response Optimization

Initial Product Backlog

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Product Backlog

User Story 1

User Story:

As a CIO, I want to see a dashboard of total incident volume over time in order to understand workload trends and resource demand.

Description:

The system will provide a time-series visualization displaying incident volume by day, week, and month.

Priority: High

Priority Justification:

Incident volume trends are foundational for all other analysis. This delivers immediate stakeholder value by revealing workload patterns and supporting staffing and budgeting decisions. It also enables subsequent diagnostic analysis.

User Story 2

User Story:

As a CIO, I want to view average and median resolution time by incident priority in order to evaluate performance against expectations.

Description:

The dashboard will display resolution time metrics segmented by priority levels.

Priority: High

Priority Justification:

Resolution time directly impacts operational efficiency and cost. This metric aligns with the project objective to reduce internal IT spend and improve productivity.

User Story 3

User Story:

As a CIO, I want to monitor SLA adherence rates in order to ensure compliance with service commitments.

Description:

The system will calculate and visualize SLA compliance percentages and highlight violations.

Priority: High

Priority Justification:

SLA compliance has financial and reputational implications. This is a critical operational KPI and provides immediate business value.

User Story 4

User Story:

As an IT Operations Manager, I want to compare resolution times across assigned groups in order to identify performance gaps.

Description:

The dashboard will display comparative charts showing average resolution time by assigned group.

Priority: Medium

Priority Justification:

This supports diagnostic analysis and helps uncover bottlenecks but depends on the foundational metrics established in higher-priority stories.

User Story 5

User Story:

As an IT Operations Manager, I want to analyze incident categories by frequency and resolution time in order to detect recurring problem areas.

Description:

The system will visualize incident categories ranked by volume and average resolution time.

Priority: Medium

Priority Justification:

Category-level analysis provides meaningful diagnostic insight but is secondary to core performance indicators.

User Story 6

User Story:

As a Data Analyst, I want to clean and validate the ServiceNow event log data in order to ensure accurate reporting and modeling.

Description:

Data preprocessing will address missing values, duplicates, and inconsistencies before analysis.

Priority: Medium

Priority Justification:

Accurate dashboards and predictive modeling depend on reliable data. While foundational, this story supports other high-value deliverables rather than delivering direct executive insight.

User Story 7

User Story:

As a CIO, I want to visualize incident lifecycle stages (opened, assigned, resolved) in order to identify process delays.

Description:

The dashboard will display the average time spent in each lifecycle stage.

Priority: Medium

Priority Justification:

This provides additional diagnostic value but is less critical than SLA adherence and resolution time metrics.

User Story 8

User Story:

As a CIO, I want a predictive model that estimates incident resolution time based on incident attributes in order to proactively allocate resources.

Description:

The system will build and evaluate a regression or machine learning model using historical incident data.

Priority: Medium

Priority Justification:

The predictive model supports improvement objectives but depends on prior measurement and diagnostic understanding. It provides strategic value but is not foundational.

User Story 9

User Story:

As an IT Operations Manager, I want the system to recommend actionable process improvements based on modeling insights in order to increase productivity.

Description

The system will generate data-driven recommendations based on model outputs and trend analysis.

Priority: Low

Priority Justification:

Recommendations depend on dashboards and predictive modeling being complete. While valuable, this story cannot be implemented until earlier analytical work is finalized.

User Story 10

User Story:

As a CIO, I want downloadable reports of key dashboards in order to share insights with executive leadership.

Description:

The system will allow export of dashboard summaries in PDF or presentation format.

Priority: Low

Priority Justification:

Export functionality enhances usability but does not directly impact operational efficiency or core analysis.

User Story 11

User Story:

As an IT Operations Manager, I want to filter dashboards by time period, priority, and category in order to conduct customized analysis.

Description:

Interactive filtering options will allow users to adjust dashboard views dynamically.

Priority: Low

Priority Justification:

Filtering improves user experience but is not essential for delivering initial project objectives.

Team Participation Statement

All team members actively participated in the development of this product backlog. The team collaborated to analyze the Viking Software case study, identify key stakeholders, and develop user stories that reflect the functional requirements of the system. Team members worked together to refine the structure of each user story, assign priorities based on business value, and ensure the backlog follows Scrum principles and the required user story format.

Team Member Signatures

Member 1: Eric Chen

Member 2: Latifa Osman

Member 3: Nishaaj Islam